

PREET JITENDRAKUMAR PATEL

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EDUCATION

M.Tech Dual Degree (5-Year) Indian Institute of Technology, Kharagpur CGPA: 8.90 / 10 2019 - 2024

WORK EXPERIENCE

American Express, Gurgaon
Senior Data Analyst Mar 2026 - Present
Data Analyst Jun 2024 - Feb 2026
Universal Modeling Automation Suite (UMAS) Jan 2025 - Present

- Objective: Automated end-to-end ML pipeline from data ingestion to governance, cutting manual modelling effort by 70% while preserving model performance.
- Engineered dynamic variable selection using permutation importance, PSI-based stability filtering, and redundancy pruning to identify and retain the most robust predictors across datasets
 - Built a model interpretability layer using SHAP TreeExplainer, partial dependence plots (PDP), and ICE curves to ensure model behaviour aligned with business expectations and domain logic
 - Automated bias and fairness checks using adverse impact analysis scoring, local heterogeneity analysis, and conditional PDP disaggregation to detect and surface bias across protected demographic attributes
 - Developed a YAML-templated auto-documentation engine using Jinja2 with Git-based versioning to generate governance-compliant model cards, and benchmarked performance via Optuna tuning and champion-challenger testing retaining baseline performance in >92% of instances

Governance Chatbot Aug 2024 - Dec 2024

- Objective: Production RAG-based LLM chatbot for compliance policy Q&A; using LangChain and Llama 3.1 (8B), deployed on-premise, reducing policy query workload by ~90%.
- Engineered a document ingestion pipeline covering PDF and DOCX parsing, recursive chunking at 2048 tokens with 128-token overlap, metadata tagging, and ChromaDB vector storage, cutting retrieval latency by approximately 30%
 - Boosted contextual accuracy by ~15% by combining FAISS dense search with BM25 sparse search using reciprocal rank fusion (RRF) and MMR reranking, with embedding clustering to reduce chunk redundancy
 - Mitigated hallucinations using semantic similarity thresholding and context faithfulness checks, and added sliding-window semantic memory to sustain coherent multi-turn policy dialogue
 - Benchmarked the system using an LLM-as-a-judge framework alongside BLEU, ROUGE-L, and Precision@K metrics, achieving 75-80% factual accuracy on a held-out policy QA benchmark

INTERSHIPS

Data Analyst Intern - American Express, Gurgaon | Video Tagging and Capsule Video Creation May - Jul 2023

- Built and deployed a fine-tuned Wav2Vec 2.0 ASR pipeline to transcribe financial video content with precise sentence-level timestamps across a large internal corpus
- Fine-tuned Vicuna-7B-Instruct and LLaMA-2-7B using LoRA adapters with 4-bit QLoRA quantisation to produce structured domain-specific video summaries
- Optimised prompts autonomously using Autoprompt gradient-based token search and real-time feedback, benchmarking all outputs across a composite evaluation framework of 10+ quality criteria

Computer Vision Intern - Polybee, Bangalore | Tomato Phenotyping Apr - Jul 2022

- Reconstructed 3D point clouds of tomato plants from RealSense D435 stereo and depth frames using Open3D ICP registration and TSDF volumetric fusion, cutting manual data collection time by 20%
- Fine-tuned Mask R-CNN (ResNet-50 FPN, Detectron2) with random-erasing augmentation improving occluded-instance accuracy by 2%, and integrated a SORT tracker using Lucas-Kanade optical flow for temporal plant tracking

PROJECTS

Intelligent QA System Mar 2023

- Built a semantic search pipeline using sentence-BERT (all-mpnet-base-v2) embeddings indexed into a FAISS IVF-PQ structure, with TF-IDF and WordNet query expansion over 10,000 structured reports
- Integrated GPT-4 with dynamic context assembly to generate grounded, accurate responses, achieving a top-5 retrieval accuracy of approximately 87% on a 500-query evaluation set

Optimisation of Socially Responsible Investment and Portfolios using Deep Learning Jul - Aug 2022

- Scraped and preprocessed financial and ESG metrics for S&P; 500 companies via the yfinance API, engineering features across return, risk, and sustainability dimensions for downstream modelling
- Applied CAPM, Fama-French 3-factor, and Black-Litterman models to optimise asset allocation and plotted the mean-variance efficient frontier to identify risk-return optimal portfolio compositions
- Trained a multivariate BiLSTM on rolling 60-day windows to project annual equity returns and generated an optimised portfolio balancing financial performance with ESG screening criteria

COMPETITION

Smart India Hackathon - National Winner Mar - Aug 2022

- Led a 6-member IIT Kharagpur team to victory over 6,500+ competing entries by building a Sanskrit speech recognition plugin using DeepSpeech2 (CNN + BiRNN + CTC, WER 18.12%) trained on 78 hours of audio deployed via a Chrome extension

AWARDS AND RECOGNITION

American Express: SVP Star Award (Q4 2025) | Analyst of the Quarter (Q3 2025) | CFR Excellence (Q1 2025) | Analyst of the Quarter (Q4 2024)

SKILLS

ML/DL Frameworks	PyTorch, TensorFlow, Keras, Detectron2, YOLOv8, Wav2Vec 2.0, HuggingFace Transformers, LangChain, FAISS, ChromaDB, ONNX Runtime
LLM and GenAI	RAG, Prompt Engineering, Fine-tuning (LoRA, QLoRA), LLM Evaluation, LangSmith, Vector Databases, Custom GPT
Coursework	Machine Learning, Deep Learning, High Performance Scientific Computing, Data Structures and Algorithms, DBMS, Business Analytics, Financial Markets